



Advanced Binary Search 3

Binary Search on Decimals

Find a real number between L and R following certain properties.

- How many iterations?
- Precision issues?
- Ways to tackle


$$O(\log(s:s))$$

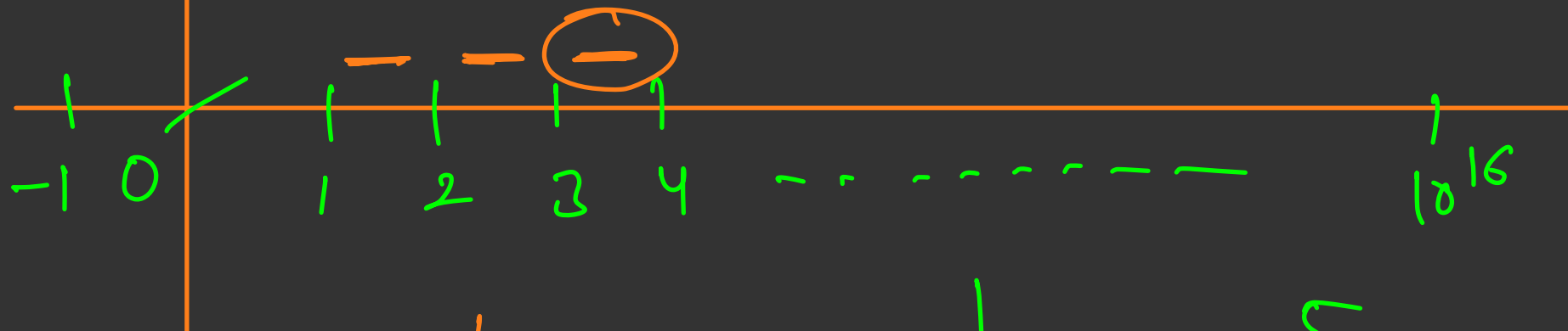


$$O\left(\log \left[\frac{\text{int } s:s}{\text{precision}} \right]\right)$$

$$\underline{\text{Sqrt}(2)} \quad \underline{10}$$

$$\underline{\text{Sqrt}(x)}$$

$$\underline{y^2 \leq x}$$



$$\underline{\underline{\frac{1}{3}}}$$

$$\underline{\underline{\frac{1}{3}}}$$

$$\frac{5}{2} = \underline{\underline{2.5}}$$



$$0.3333 \dots$$

$$\underline{\underline{0.33}}$$

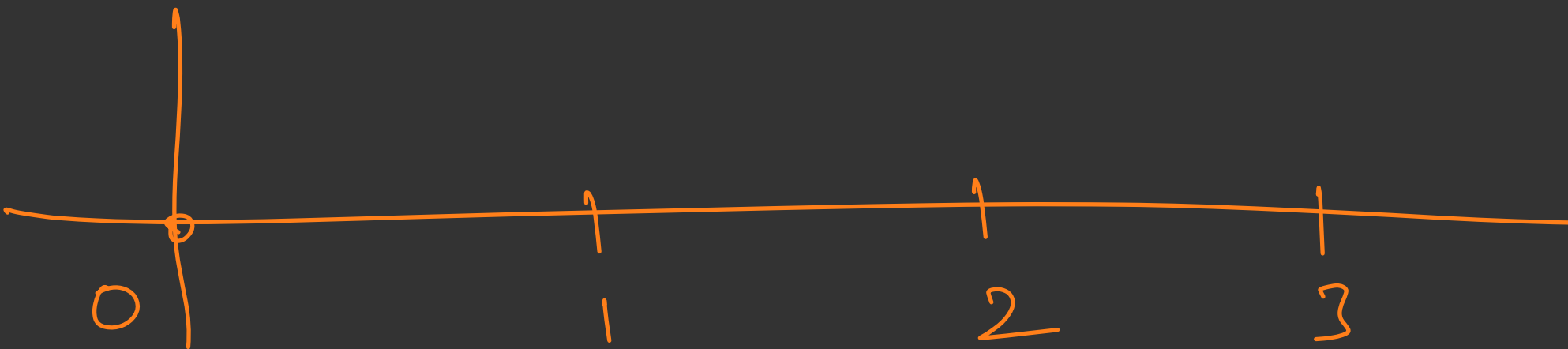
Print the answer which has a relative error of $\leq 10^{-6}$ with the original answer

Your ans $\Rightarrow X$

Original ans $\Rightarrow Y$

$$|X - Y| \leq 10^{-6}$$

$$Y + 10^{-6} \geq X \geq Y - 10^{-6}$$



1.8000001
1.0000002

←

original ans = 1.000123123

You print = 1.000123 ✓
1.000124 ✓

orig $\rightarrow$ ~~x~~

you as $\rightarrow$ y

$$|x - y| \leq 10^{-6}$$

~~_____~~

orig $\rightarrow$

1.000123123

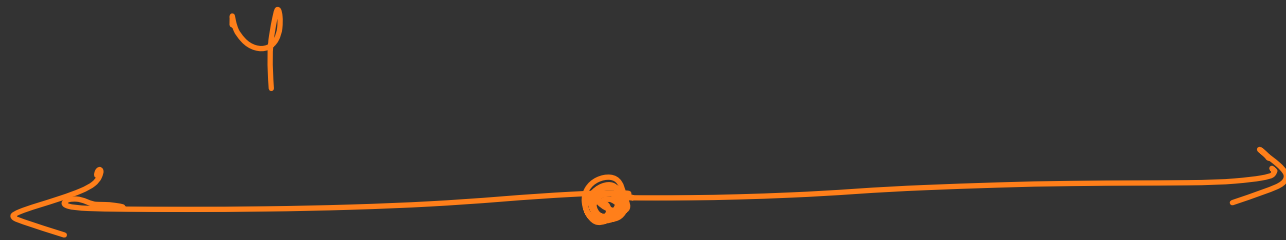
printed

1.000124

as

$\Rightarrow$

1.000122



$$x - 10^{-6} \quad \longleftarrow \quad x \quad \longrightarrow \quad x + 10^{-6}$$

$$\underline{\underline{10^{-6}}}$$

$$x = 1.000123123 \quad \leftarrow$$

$$y = 1.000123 \quad \checkmark$$

$$1.000124 \quad \checkmark$$

$$1.000122 \quad \times \quad \leftarrow$$

$$x - y = 1.000123123$$

$$- 1.000122$$

$$= 0.00000123 = > \underline{\underline{10^{-6}}}$$

$$Y = 1.800124000$$

$$X = 1.000123123$$

$$0.0000000877$$

$$X = 1.800123999$$

$$\underline{y^2 \leq \boxed{\times}}$$

Act ans = 1 0 0 0 1 2 3 1 2 3

1 0 0 0 1 2 2
 [1 0 0 0 1 2 3]
 1 0 0 0 1 2 4

↓ ↓ ↓
 T T T T (T) (F) F F F F F

Size of the search space

0 $\longrightarrow$ 10^{16}

8 decimal places

10^{16} integers $\times 10^8$

Search space = $\frac{\text{Integer search space}}{\text{precision}}$

10^{16} int

8 decimal

$$\rightarrow \frac{10^{16}}{10^{-8}} = 10^{24}$$

0 $\xrightarrow{\hspace{10em}}$ 10^{16}
8 decimal places

$$\underline{10^{16} \times 10^8}$$

$$10^{16} + 10^8$$

is binary searching on 10^{24}
ans

same

first binary search on 10^{16}

and then find on 10^8

$$T.C \rightarrow O(\log(S \cdot S))$$

$$S \cdot S = 10^{24}$$

$$\rightarrow O(\log(10^{24}))$$

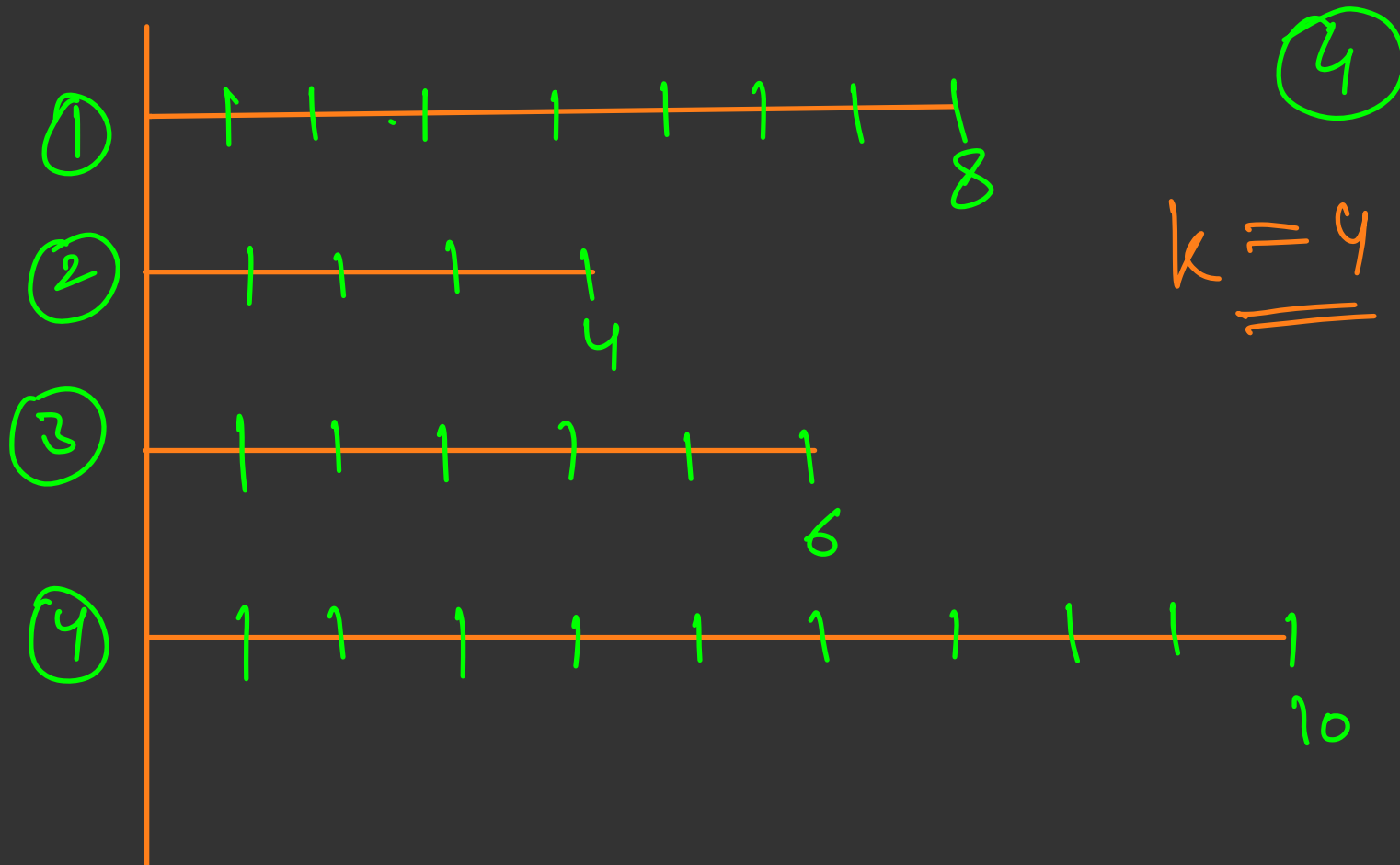
$$O(\log(10^{16})) + O(\log(10^8))$$

$$\log(a) + \log(b) = \log(ab)$$

Problem 1: Ropes

n ropes $\rightarrow$ 1 to 10^7

(k)



Binary search on the maximum length

$f(x) \rightarrow$ T if we can cut out $\geq k$ pieces of x length

$\searrow$ F o/w

$\min = 0, \quad \max = 10^7$

$$\text{max} = 10^7$$

$$n \rightarrow 10^4$$

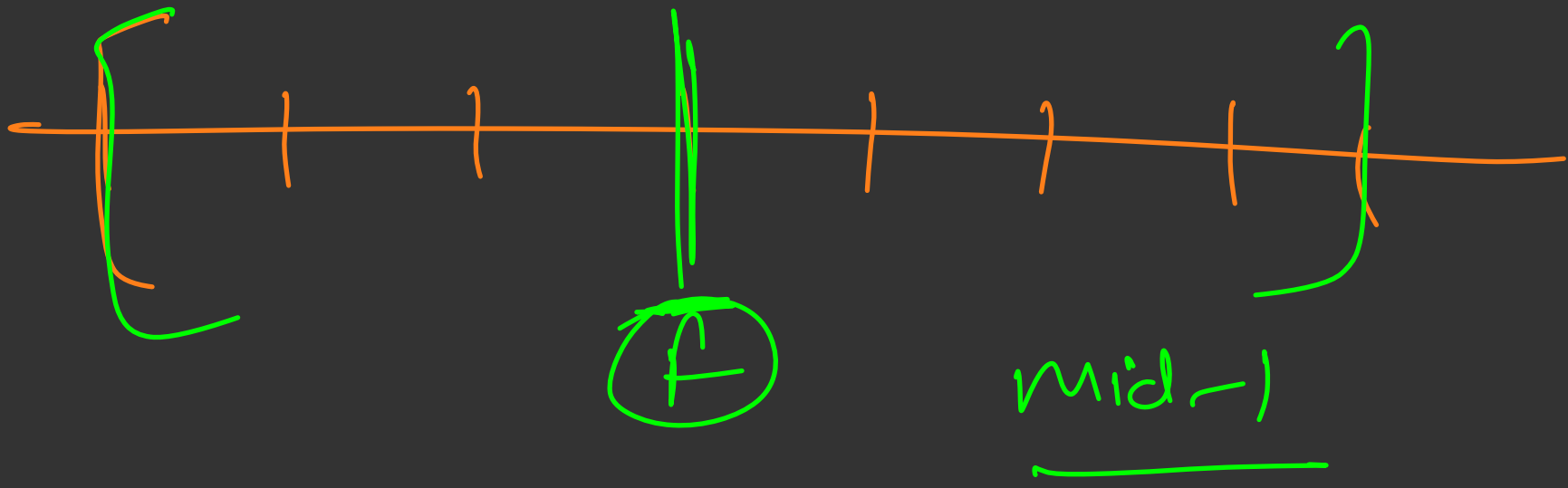
$$1 \leq q_i \leq 10^7$$

$$1 \leq k \leq 10^9$$

$$\min \geq 0,$$

$$m_{\text{an}} \approx 10^{18}$$

1 2



T T T T T f f f f f f f

error

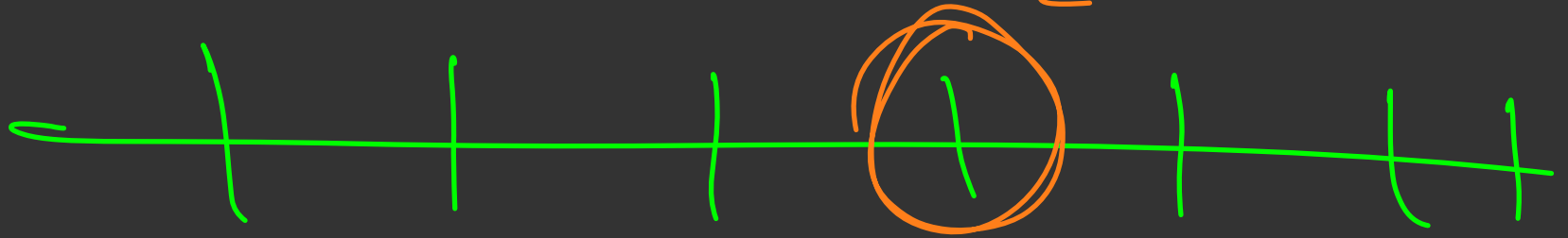
10^{-6}

mid - err
mid + err

mid - 1

mid - 10^{-6}

while ($l \leq u$) {
 $mid \neq (l+u)/2$



}

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A hand-drawn diagram of a neural network layer with 10 nodes. The top row shows weights: -1, 1, 1, 1, 1, 1, 1, 1, 1, 1. The bottom row shows bias: 1, 1, 1, 1, 1, 1, 1, 1, 1, 1. The 7th node is highlighted with a green circle and a red arrow pointing to it from the left.

0.001 0.002 0.003 0.004 0.005 0.006 0.007 0.008 0.009 0.010

$$\underline{\underline{10^{-6}}}, 10^{-7}$$
$$\begin{array}{r} 0'00061 \\ 0'00062 \\ \hline 0'00063 \\ \vdots \end{array}$$

$f(n) \rightarrow \top$ if we can cut k

$$\boxed{a_i / n}$$

pieces from all ropes

~~of length n~~ of length n

_____ a_1

_____ a_2

_____ a_3

... $a_1 + a_2 + a_3 + \dots \geq k$

```

bool checker(double x, vector<int>& arr, int k){
    long long total = 0; ← int
    for(auto i : arr){
        total += i / x;  = int
        if(total >= k){
            return true;
        }
    }
    return false;
}

void solve(){
    int n, k;
    cin >> n >> k;
    vector<int> arr(n);
    for(int i = 0; i < n; i++){
        cin >> arr[i];
    }
    double left = 0, right = 1e7, error = 1e-7;
    double ans = 0;
    while(right - left > error){
        double mid = (left + right) / 2;
        if(checker(mid, arr, k)){
            ans = max(ans, mid);
            left = mid + error;
        } else {
            right = mid - error;
        }
    }
    cout << ans
    // cout << setprecision(7) << fixed << ans << endl;
}

```

$O(\log(S.S) \times O(n))$
 $O(\log(\frac{10^7}{10^7}) \times O(n))$
 $= O(\log(1)) \times O(n)$
 $= 0 \times O(n)$
 $= 0$

Any problems with this code?

~~TTTTTTTTTT~~

$(45) \times 10^4$

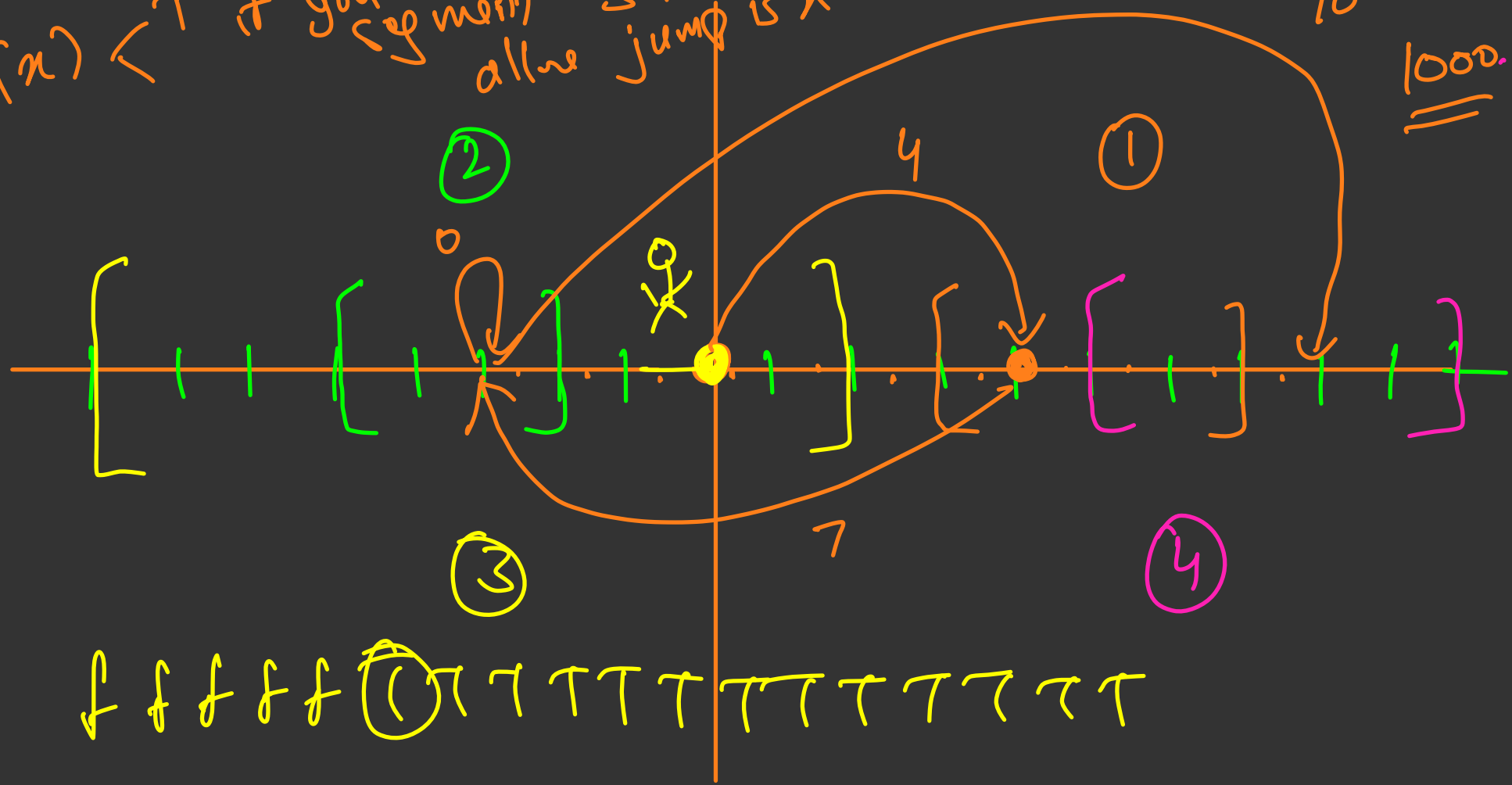
Problem 2: Jumping Through Segment

[CP-31 1400: P3]

$f(n) < T$ if you can reach with
 segment $\leq n$ all the jump $\leq n$

(10) (11), (12), (13)
 10

1000.



f f f f f (1) T T T T T T T T T T T

